Week 2 optimization and search

1. What is search landscape? What is the z-axis?

A search landscape is a plane that shows the best values for two features. It represent the space of solutions. The z-axis represent a function of how good a combination of features are.

2. What is the goal of optimalization algorithms?

The goal of optimalization is to find the best values for features. The goal is to as efficiently as possible search the landscape for the best solution.

3. What is the difference between global and local optima?

Local optima is the best solution between neighbours, but the global optima is the best solution over all (in the landscape)

4. What is the difference between continuous and discrete optimization? Name some discrete optimalization methods and name a continuous optimalization method.

Continuous optimalization is the optimalization where we do not have specific values of features to test inbetween. Discrete is when we have a set of solution to test between.

Continuous is when we have to search in a continuous spectrum. Discrete is when we ha a selected number

5. What is exhaustive search? When is it beneficial to use es (what type of graph)? Is it guaranteed to find the best solution?

Exhaustive search is when we test all solution to find the best. It is the stupidest solution, because we have to test all of the solution, which takes a long time. It is however always guaranteed to find the best solution. It is beneficial to use exhaustive search for small datasets, or when there is only one solution. It is also beneficial to use if the points are very random.

6. What is greedy search?

Only generates and evaluate one solution. Greedy search makes several locally optimal choices.

7. What is hill climbing? When do we stop?

Hill climbing is when we take a random solution, and test this solution with its neighbour. If the neighbour is better, then this becomes the current best solution. We test the current solution with its neighbour until we do not find a better neighbour.

8. What is exploitation and exploration? What category is exhaustive search and hill climbing?

Exploration is when we test a random solution. Exploitation is when we make the current solution better. Exhaustive search is exploration. Hill climbing is exploitation.

9. What is simulated annealing?

Simulated annealing is a optimalization algorithm that has a temperature that decides how much exploration and exploitation. The temperature is set high at first, that means high exploration and low exploitation. Then the temperature gradually becomes lower, with lower exploration and higher exploitation.

10. What is gradient ascent/descent?

Gradient ascent/descent is when we start with a random solution and look at the gradient (derivative) to go in the direction of the global optima. We use a learning rate or gamma to choose how much we change the solution. Low gamma can take longer time to find the best solution, big gamma can risk overshooting.

11. Gradient ascent and hill climbing are quite similar, and they are based almost exclusively on exploitation. Can you think of any additions to these algorithms in order to do more exploration?

We can add more randomness when we test it. For example run the algorithm several times with random starting solutions.

Not always choose the best solution